Offense and Defense (Overall Summary)

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Outline

- How are web attacks carried out?
- How to defend against them?
  - Web Application Firewalls
- Overall Best Practices to follow (Generic)

Attack Process

How are attacks typically carried out?

- Reconnaissance (Information Gathering): Gather information about the target web application
  - Technologies used (e.g., server software, frameworks, libraries) using tools like Wappalyzer or WhatWeb
  - Cross-reference with CVE databases to find known vulnerabilities (e.g., outdated versions)
  - Use ZAP to map the site structure
    - Identify all endpoints, hidden pages, and forms using spiders and and manual exploration

- Scanning and Enumeration: Actively probe the web application to map its structure and identify specific vulnerabilities
  - Automated vulnerability scanners like OWASP ZAP, Burp Suite, Nikto, or Acunetix
  - Examine input fields and parameters for injection points
  - Analyze cookies and session tokens
  - Identify misconfigurations, outdated software, or missing security patches
  - Enumerate user accounts or roles via error messages or predictable pattern

- Gaining Access (Exploitation): Exploit vulnerabilities to gain unauthorized access to the web application
  - Injection Attacks: SQL injection, Command injection, XML external entity (XXE)
  - Cross-Site Scripting (XSS): Injecting malicious scripts to run in users' browsers
  - Broken Authentication: Exploiting weak login mechanisms, session fixation, or credential stuffing
  - Insecure File Uploads: Uploading malicious files or web shells
  - Path Traversal: Accessing sensitive files on the server
  - Cross-Site Request Forgery (CSRF): Forcing logged-in users to execute malicious actions etc

- Privilege Escalation: Elevate access rights to gain more control over the application or underlying infrastructure
  - Exploiting insecure direct object references (IDOR) to access restricted resources
  - Leveraging insecure role-based access control (RBAC) to access admin-level functionality
  - Hijacking sessions or tokens for privileged accounts
  - Exploiting vulnerabilities in the web server, database, or operating system

- Lateral Movement: Move within the network or exploit other related systems connected to the compromised application
  - Accessing backend databases using SQL injection
  - Interacting with APIs to access additional functionality or data
  - Using the compromised application as a pivot point to attack internal systems or other applications
    - Exploiting trust relationships between systems

- Exfiltration: Steal sensitive data or assets from the application
  - Databases (e.g., extracting user credentials, PII, or financial data)
  - Session cookies or tokens for further impersonation
  - Sensitive files (e.g., configuration files, logs, or source code)
  - API keys and credentials

- Maintaining Access (Persistence): Maintain control over the compromised web application for further attacks or data exfiltration
  - Installing web shells for remote access (e.g., PHP-based shells)
  - Leveraging backdoors in source code or server configurations
  - Adding rogue admin accounts or API keys

- Covering Tracks: Erase evidence of the attack to avoid detection
  - Delete logs or modifying log entries to hide malicious activities
  - Clean up uploaded web shells or malicious files
  - Obfuscate payloads to make analysis harder
  - Use encrypted or encoded communication for data exfiltration.

Defense

- Reconnaissance:
  - Limit public information exposure
  - Use tools like WAFs (Web Application Firewalls) to detect and block reconnaissance activities
- Scanning and Enumeration:
  - Monitor unusual scanning activity in server logs
  - Harden application endpoints and implement rate limiting

- Exploitation:
  - Regularly patch software and frameworks
  - Perform security assessments (e.g., penetration testing, SAST, DAST) and fix issues
  - Enforce best security coding practices ; data validation
- Privilege Escalation:
  - Implement least-privilege access controls
  - Use role-based access control (RBAC) with proper testing
- Lateral Movement:
  - Segment the network to limit attack propagation
  - Restrict access to internal systems via firewalls or proxies and also via authentication

- Exfiltration:
  - Encrypt sensitive data in transit and at rest
  - Monitor unusual outbound traffic patterns
- Persistence:
  - Regularly audit application configurations and accounts.
  - Detect and remove rogue files or accounts
- Covering Tracks:
  - Use logging and monitoring solutions with tamper-proof mechanisms
  - Employ forensic tools to analyze suspicious activity

Web Application Firewalls

- Firewall that protects web applications by filtering and monitoring **HTTP traffic** from Internet
  - Operates at the application layer
  - Inspects both incoming and outgoing traffic
- Can identify and block a variety of attacks
  - SQL injection, cross-site scripting (XSS), cross-site request forgery (CSRF), DDOS etc
- Can help organizations meet regulatory compliance requirements
  - PCI DSS, HIPAA, and GDPR etc

Types of WAFs

- Hardware-Based: Physical devices installed at the network perimeter
  - High performance and low latency but costly
  - Complex deployment and configuration
    - Suitable for large enterprises, data centers, regulated industries!
    - Imperva SecureSphere, Barracuda Web Application Firewall, Citrix Netscaler Application Firewall etc

- Network-Based: Hardware or virtual appliance installed at network edge
  - Similar to hardware WAF if not virtual
    - "Hardware WAF" refers to the physical appliance form factor
    - "Network-based WAF" focuses on its placement within the network architecture
  - Suitable for handling high volumes of traffic
  - Requires specialized network knowledge for management

- Host-Based: Software modules or agents installed on the web server
  - Deep integration with web application allows high customization
  - Can Consume server resources, affecting performance
  - Requires management and maintenance on each server
  - E.g. Ngnix ModSecurity, Owasp Coraza (open source)

- Cloud-Based: Delivered as a cloud service by third-party providers
  - Not your headache, infrastructure wise!
    - Incoming traffic destined for on-premises web servers is routed through cloud-based WAF
    - Achieved through DNS changes or a reverse proxy
  - Easily scales to handle large traffic volumes
  - Potential additional latency depending on provider proximity
  - E.g. AWS WAF, Cloudflare WAF, Akamai Kona Site Defender

- Hybrid WAF: Combination of network-based, host-based, and cloud-based WAFs
  - Different parts of the application infrastructure protected differently
  - Increased complexity in configuration and management

Models

Based on approach to traffic filtering and rule enforcement

- Positive Security Model
  - "Allowlist": defines what is allowed and blocks everything else by default
    - Only known, trusted, and explicitly defined inputs are permitted
  - Provides higher security assurance but high maintenance overhead
    - Requires continuous updating of allowlists
    - Potential for Blocking Legitimate Traffic
  - Works in environments where security is prioritized over convenience :-)

- Negative Security Model
  - "Blocklist": defines what is not allowed, permitting all other traffic by default
  - Relies on patterns, signatures, and rules to identify and block known malicious traffic
  - Easier to deploy initially but has false negatives
    - May miss new or unknown threats that aren't covered by the existing rules and signatures
  - Reactive Approach: Requires constant updating to keep up with emerging threats
  - Works in organizations needing a more flexible and adaptable security solution

- Mixed Security Model: Combines elements of both positive and negative security models
  - Uses allowlists for known, critical inputs
  - Uses blocklists to filter out known malicious patterns
  - A balanced approach that leverages the strengths of both models
  - More complex to configure and manage due to the dual nature of the rules
  - Complex applications where both known good inputs and dynamic traffic need to be managed

Modes

Based on how WAF interacts with traffic

- Passive Mode: WAF monitors and analyzes traffic without actively blocking or modifying it
  - Generates alerts for suspicious or malicious activities but allows all traffic to pass through
  - Zero Latency since it doesn't alter traffic
  - Useful during initial deployment phase to understand the traffic patterns
    - Can fine-tune the rules before moving to an active mode
  - Drawbacks:
    - Does not block malicious traffic, so application remains vulnerable
    - Requiring manual intervention to respond to threats

- Active Mode: Actively filters, blocks, and modifies traffic based on predefined rules and policies
  - Can prevent malicious traffic in real-time
  - Risk of False Positives: May block legitimate traffic if the rules are too strict or not properly tuned
  - Can introduce some latency due to the inspection and filtering processes
  - Requires careful configuration and continuous tuning to balance security and accessibility

Deployment Models

- Reverse Proxy Deployment: WAF acts as an intermediary between clients and web server
  - Breaks the connection in between client server
    - HTTPS → WAF needs access to private keys
  - Requires changes in DNS settings or configurations to direct traffic through the WAF
  - Adds an extra hop in the traffic flow, which can introduce latency
  - Single Point of Failure: If WAF goes down, can disrupt access to the web application

- Inline Deployment: WAF placed directly in the path of traffic between the client and the web server
  - Can break connection too!
  - Any issues with the WAF can cause downtime for web application

- Out-of-Band Deployment (Passive/Monitor Mode): WAF monitors a copy of the traffic through a network tap or port mirroring
  - HTTPS traffic → SSL offloading has to happen in some front-end device
  - No Latency
  - No Risk of Disruption
  - No Real-Time Blocking
    - Can only alert on malicious traffic, not block it in real-time
    - Requires manual intervention to mitigate threats

- Transparent Bridge Mode: Operates as a bridge at the network layer (Layer 2)
  - Rather than as a traditional proxy at the application layer (Layer 7)
  - Useful in scenarios where existing network setups cannot accommodate IP or DNS changes
  - Complex Configuration: Can be more complex to set up and maintain
  - Potentially Limited Features: Some advanced features may not be available in transparent mode

Internal Mechanism

- Signature-Based Detection:
  - Employ predefined signatures or patterns of known attacks
  - When incoming traffic matches a known attack signature, block or log the request
  - For XSS attacks, sample signature in URL
    - <\s*script[^>]*>(.*?)<\s*/\s*script\s*> (Matches any <script> tag with any content inside it and closes with a </script> tag.)
  - For SQL injection, sample signature in URLs
    - '(\s|;|--)|(\s|;|--)'[^'] (use of the single quote character in unexpected places

- Anomaly-Based Detection:
  - Establish a baseline of normal behavior for web traffic
    - Request rates, HTTP methods, sizes of payloads, URL patterns
    - Flag requests or responses that deviate significantly from this baseline
    - Can help detect new or zero-day attacks that may not have a known signature

- IP Reputation and **Rate Limiting**
  - WAFs may maintain lists of known malicious IP addresses (IP reputation lists)
  - Use rate limiting to protect against brute force attacks, DDoS
- Integration with external Threat Intelligence
  - Dynamically update detection rules and signatures based on emerging threats

- Logging and Reporting
  - Log all detected events, including blocked requests, anomalies, and other security incidents
  - Useful for forensic analysis, compliance auditing, and improving effectiveness of policies
- Response Actions
  - block the request
  - redirect to a CAPTCHA challenge page
  - alerting administrators, or modifying the request or response to mitigate the threat

Rate Limiting

- Count Requests: Tracks the number of requests in a specific time frame
  - Particular client (often identified by IP address or API key)
  - Type of request
- Thresholds: Maximum number of requests allowed in a given period
- Blocking or Throttling: Exceed the rate limit → block or slow down the response rate (throttling)
- Response Codes: Clients informed of rate limiting via HTTP status codes such as 429 (Too Many Requests).

- Balancing Limits:
  - Setting limits too low can disrupt legitimate users
  - Setting them too high might not effectively prevent attacks
- Distributed Attacks:
  - Rate limiting per client not very effective against distributed attacks (DDoS)

Limitations

- Cost: Hardware WAFs and advanced cloud-based solutions can be expensive
- Complexity
  - Configuration and Tuning requires expertise to avoid false positives and false negatives
    - False Positives: Legitimate traffic may be blocked if the WAF rules are too strict
    - False Negatives: Malicious traffic may go undetected if WAF rules are too lenient or not updated

- Performance Impact
  - Inspecting and filtering traffic can introduce latency, potentially affecting application performance
  - Can consume significant network and server resources, particularly in high-traffic environments
- Effectiveness of a WAF heavily depends on its configuration
  - Poorly configured WAFs can provide a false sense of security!

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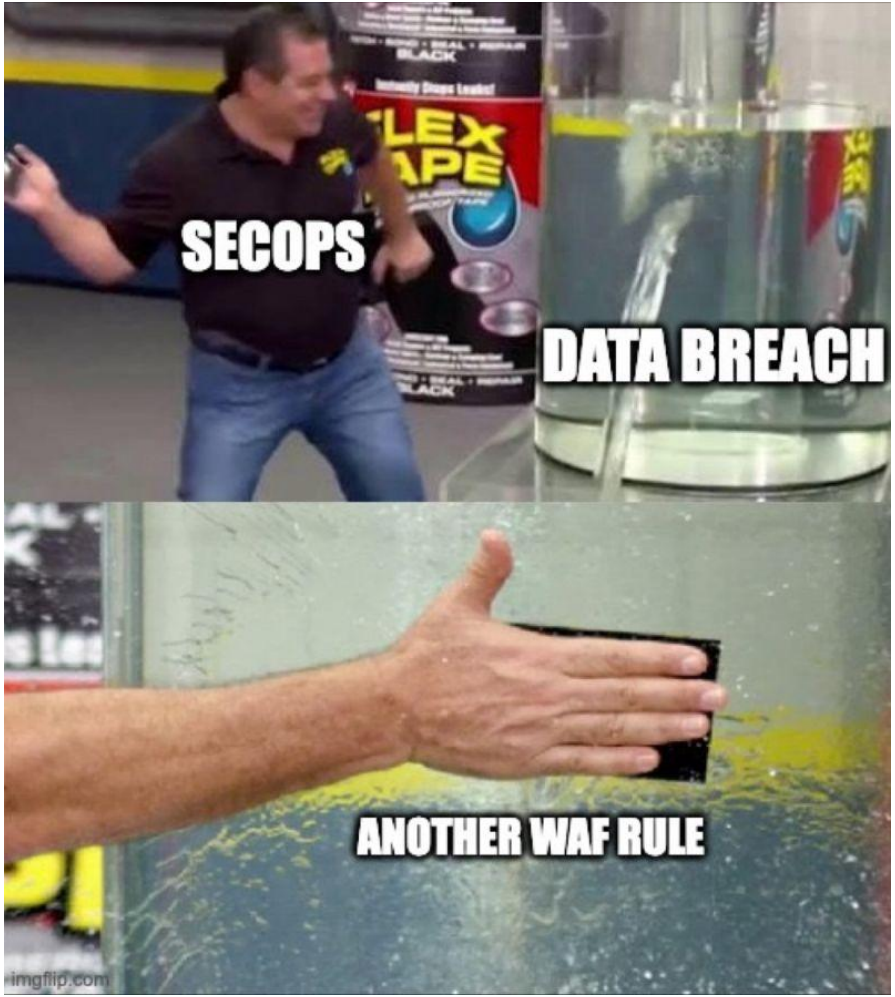

Best Practices

Security is Economics

Adi Shamir: “ There are no secure systems, only degrees of insecurity”

- No system is 100% secure
- Tradeoff between cost and security
  - $10,000 WAF to protect $1000 worth of secrets?
- Focus on securing the weakest link
  - Screen door and main door locked with expensive deadbolt

Least Privilege

- Give only minimum required privileges to get job done
  - Web applications can only read or write to specific directories/tables necessary for their operation
  - Limit API keys and tokens to specific actions and scopes required by the application
  - Implement roles with specific permissions tailored to the needs of different types of users.
  - Configure firewalls to allow only necessary traffic to and from the web server

Use fail-safe defaults

- Access to resources should be denied by default
- Permissions should be explicitly granted based on the principle of least privilege
  - Configure firewalls to block all traffic by default and only allow required traffic through rules
  - Default passwords should be strong and unique
  - Default settings should not expose any unnecessary services or data
  - Access to sensitive areas of the application requires authentication by default
  - Implement strict CSPs that disallow potentially dangerous actions by default,
  - Use whitelisting (allowing only known good inputs) rather than blacklisting (blocking known bad inputs) for form fields and user inputs

Separation of responsibility

- Split up responsibility, no one person has complete power
  - Nuclear missile launch has two launch officers
  - Purchase committee to evaluate purchases
- Separate roles for developers, testers, and administrators; Separate roles for detecting, investigating, and resolving security incidents
  - Each group has access only to the resources necessary for their tasks
- Independent auditors to review security practices and compliance regularly

Defense in depth

- Use multiple redundant protections
  - Supandi Joke
- All need to be breached before system is endangered
- Web Applications
  - Implement secure coding standards and practices to minimize vulnerabilities during development lifecycle
  - Deploy WAFs to filter and monitor HTTP traffic between a web application and Internet
  - Ensure all user inputs are validated and sanitized
  - Use Secure server configuration (CSP, .htaccess etc)
  - Implement MFA to require multiple forms of verification

Psychological acceptability/Human Oriented Design

- Users need to buy into the security model
  - E.g. firewall blocking leads to tunnels; Passwords strength leads to post-its
- Implement single sign-on (SSO) solutions to allow users to log in once and gain access to multiple systems
- Make it easy for users to change their passwords or understand the status of their security settings

- If a password is rejected, explain why and offer suggestions for creating a stronger password
- Reduce authentication requirements when users log in from familiar devices or locations but increase them when anomalies are detected!
- Automate software updates and security patches to ensure systems are up to date

- Design should be easy to use for ordinary people
  - NSA’s encryption key example
    - Key is stored in a physical token which looks like a door key
    - Very intuitive even for people with no training in crypto
- Use visual cues like color coding (green for safe, red for danger) to communicate security status
- If a user encounters a security warning, offer a brief explanation and clear steps to resolve it
- Provide clear and easy-to-use channels for reporting security incidents or suspicious activities

Ensure Complete Mediation

- Every access to a system's resources is checked for appropriate permissions and access rights
  - Implement centralized mechanisms for access control where permissions and policies are enforced consistently
  - Require users to re-enter their passwords when accessing high-risk areas of the application
  - Enforce access controls on files, databases, and other resources at every access attempt
  - Validate all input parameters every time they are received
  - Log all access attempts and security checks to maintain an audit trail

Detect if you can't prevent

- Detect; if possible provide a way to recover and identify the attacker
  - Example in hardware setting
    - Type 3 Tamper-resistance: make intrusion difficult, usually by employing hardened casing
    - Type 2 Tamper-evident: make intrusion attempts evident to subsequent viewers
      - Often by employing seals which must be broken during intrusion
    - Type 1 Tamper-responsive: detect intrusion attempt and destroy the contents
    - Type 1 most expensive, type 3 more expensive than type 2

- Implement continuous monitoring of networks, systems, and applications to detect anomalies and potential security incidents in real-time
- Incorporate threat intelligence feeds to stay updated on the latest threats and attack techniques

Don’t rely on security through obscurity

- Keeping design, algorithm, code, URLs secret is brittle
- Don’t trust such claims

Design security in from start

- Fitting in security later is a very tough task (as seen in Internet)
- Backward compatibility will make things insecure
- Conservative Design: Design against the worst security failure under assumptions favourable to the attacker
  - Kerkhoff's principle: Attacker knows all internal details
- Proactive study of attacks: Attempt to break your own system before others do it

Know your threat model

- Assumptions originally made may not be valid
  - E.g. design of Internet
- Identify Assets:
  - Types of data your web application handles (personal, IP or credentials etc)
  - Determine critical components of your system (databases, servers, APIs, and third-party integrations)
- Determine Potential Adversaries:
  - External Attackers, Insiders, Contractors, Competitors

- Identify Threats and Attack Vectors:
  - Common Attacks: SQLi, XSS, CSRF, DDos
  - Advanced Persistent Threats (APTs): more sophisticated threats that may involve long-term, targeted attacks aimed at specific high-value assets
  - Supply Chain Attacks: risks associated with third-party components and services
- Existing Controls: Evaluate the security controls currently in place
  - Firewalls, encryption, authentication mechanisms, secure server configuration etc
- Gaps and Weaknesses: Identify any gaps or weaknesses in existing security measures

- Assess Likelihood and Impact:
  - Estimate likelihood of each identified threat occurring based on factors like past incidents, industry trends, and your specific environment
  - Impact: Evaluate the potential impact of each threat on your organization
    - Data loss, financial damage, reputational harm, and legal consequences.
- Risk Assessment: Perform a risk assessment to prioritize threats based on their likelihood and potential impact
- Resource Allocation: Allocate resources and efforts proportionally to the severity and priority of the threats identified

- Develop Mitigation Strategies:
  - Preventive Measures: Implement measures to prevent threats
    - Secure coding practices, regular security training, and robust access controls
  - Detective Measures: detect threats through continuous monitoring, logging, and intrusion detection systems
  - Responsive Measures: Prepare for incident response with well-defined procedures, backup strategies, and disaster recovery plans

- Iterate and Update:
  - Regular Review: Regularly review and update your threat model
  - Testing and Validation: Continuously test and validate your security measures
    - penetration testing, and red teaming exercises
- Communication and Awareness:
  - Involve key stakeholders, including management, IT staff, and end-users
- Documentation: Document your threat model

Summary

- Web Application attack process involves a bunch of steps
- Covered defense in each step and also WAFs
  - WAFs: many types, modes, deployment models possible
    - Protect against many attacks
    - Based on combination of signature and anomaly detection
- Overall best practices (adapted to web applications)

Reference

- Generic kill chain: https://www.geeksforgeeks.org/cyber-kill-chain/
- WAFs: https://www.vmware.com/topics/web-application-firewall
- Best Practices: http://inst.eecs.berkeley.edu/~cs161/sp16/notes/2.1.principles.pdf